The Discrete Fourier Transform (DFT):

- The DFT is a computational tool used to analyze the frequency content of *finite-length* discrete-time signals.
- Our DFT discussion will include:
  1. definitions of the DFT and IDFT,
  2. interpretations of the DFT:
     (a) sampled DTFT,
     (b) discrete Fourier series,
  3. properties of the DFT, including circular convolution,
  4. use of DFT for spectral analysis,
  5. fast computation of the DFT via the FFT,
  6. matrix/vector formulations.

DFT definitions:

- For an N-length signal $\{x[n]\}_{n=0}^{N-1}$, the N-point DFT is

$$X[k] \;=\; \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}, \qquad k = 0 \ldots N-1$$

and the corresponding N-point IDFT is

$$x[n] \;=\; \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}kn}, \qquad n = 0 \ldots N-1.$$

- Note:
  - the signal has finite duration N,
  - the frequency domain is discrete and length-N.

DFT Interpretation # 1 — sampled DTFT:

Consider the finite-length signal $\{x[n]\}_{n=0}^{N-1}$. Defining $\{x[n]\}_{n=-\infty}^{\infty}$ via zero extension, the DTFT becomes

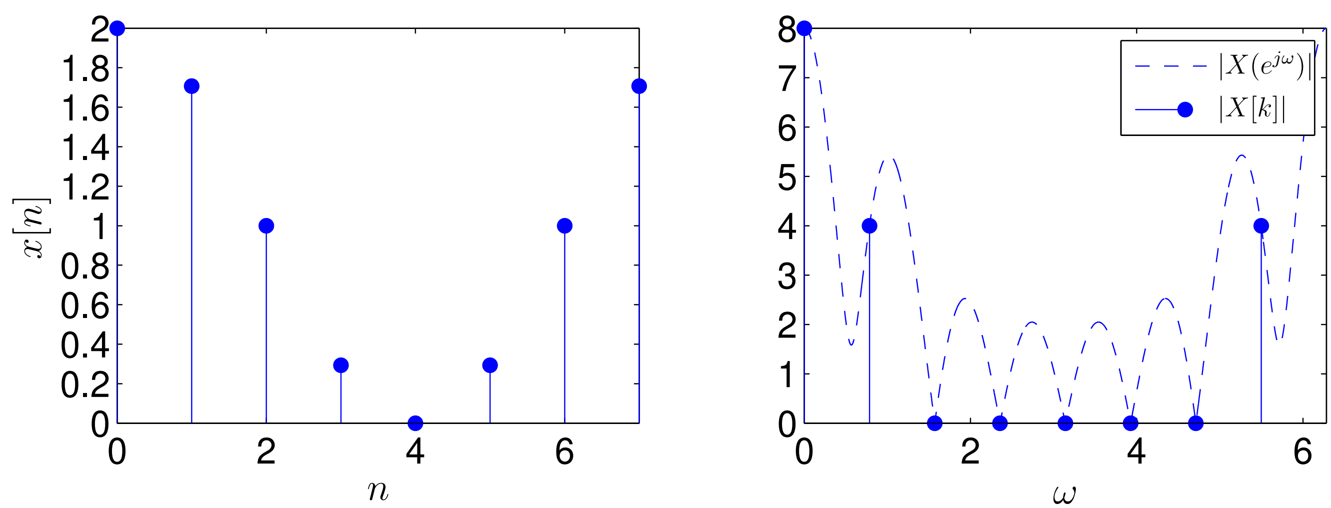
$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n=0}^{N-1} x[n]e^{-j\omega n}.$$

Meanwhile, the N-DFT of the non-zero samples is

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn} = X(e^{j\omega})\Big|_{\omega=\frac{2\pi}{N}k}$$

$$\text{for } k = 0 \ldots N-1.$$

Thus, the N-DFT returns $\frac{2\pi}{N}$-spaced samples of the (zero-extended) DTFT. Example for $N = 8$:



Notice that we've uncovered a new concept: *sampling in the frequency domain!*

The DFT and zero-padding:

Again consider a finite-length $\{x[n]\}_{n=0}^{N-1}$ and $\{x[n]\}_{n=-\infty}^{\infty}$ defined via zero-extension.

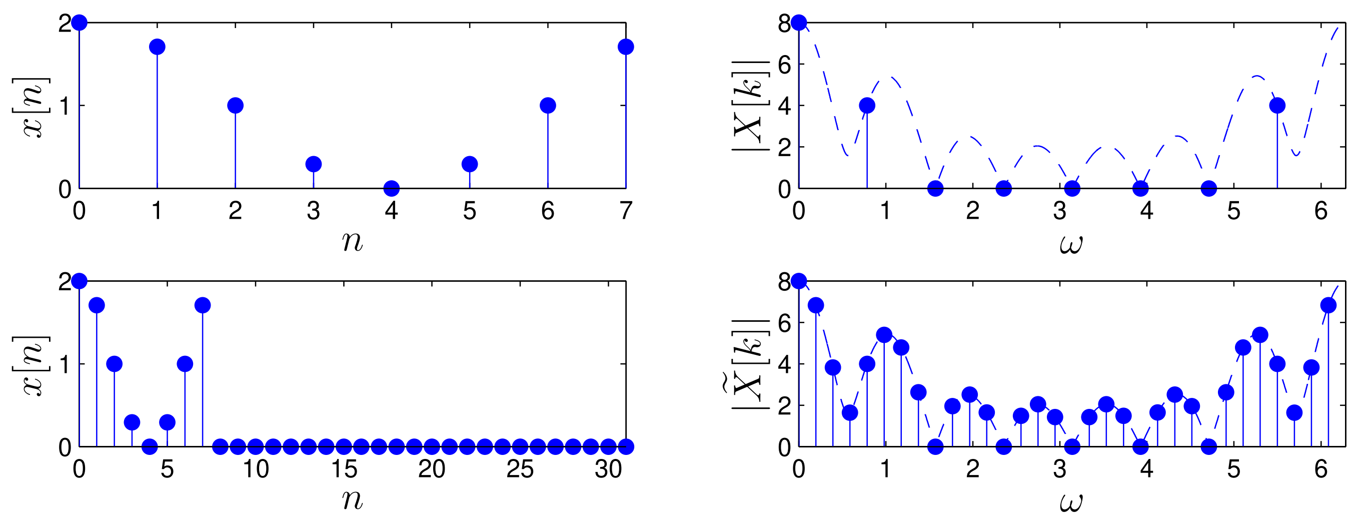
But now take the M-length DFT, where $M > N$:

$$\begin{aligned}\widetilde{X}[k] &= \sum_{n=0}^{M-1} x[n]e^{-j\frac{2\pi}{M}kn} = \sum_{n=-\infty}^{\infty} x[n]e^{-j\frac{2\pi}{M}kn} \\ &= X(e^{j\omega})\big|_{\omega=\frac{2\pi}{M}k}.\end{aligned}$$

The M-length DFT samples the DTFT at the interval $\frac{2\pi}{M}$.

The process of appending zeros to a finite-length signal is called *zero padding*.

Thus, zero-padding increases the rate at which the DFT samples the DTFT!



DFT Interpretation #2 — discrete-time FS:

Say $\widetilde{x}(t)$ is periodic with P-second period, and $x(t)$ contains only the first period of $\widetilde{x}(t)$. Then, from Fourier series,

$$x(t) = \frac{1}{P} \sum_{k=-\infty}^{\infty} P X_k e^{j\frac{2\pi}{P}kt}, \quad t \in [0, P)$$

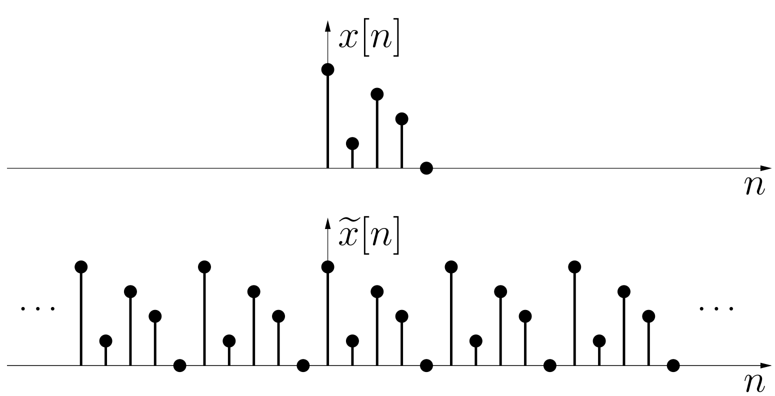
$$P X_k = \int_0^P x(t) e^{-j\frac{2\pi}{P}kt} dt = X_{\mathsf{c}}(j\Omega)\Big|_{\Omega=\frac{2\pi}{P}k}$$

Say $\widetilde{x}[n]$ is periodic with N-sample period, and $x[n]$ contains only the first period of $\widetilde{x}[n]$. Then, from the DFT,

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}kn}, \quad n = 0 \ldots N-1$$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn} = X(e^{j\omega})\Big|_{\omega=\frac{2\pi}{N}k}$$

The similarity suggests that one can interpret the DFT coefs $\{X[k]\}_{k=0}^{N-1}$ as the coefs of a "discrete-time Fourier series."



Note: zero-extension $\rightsquigarrow$ DTFT; periodic extension $\rightsquigarrow$ FS.

DFT interpretation #3 — basis expansion:

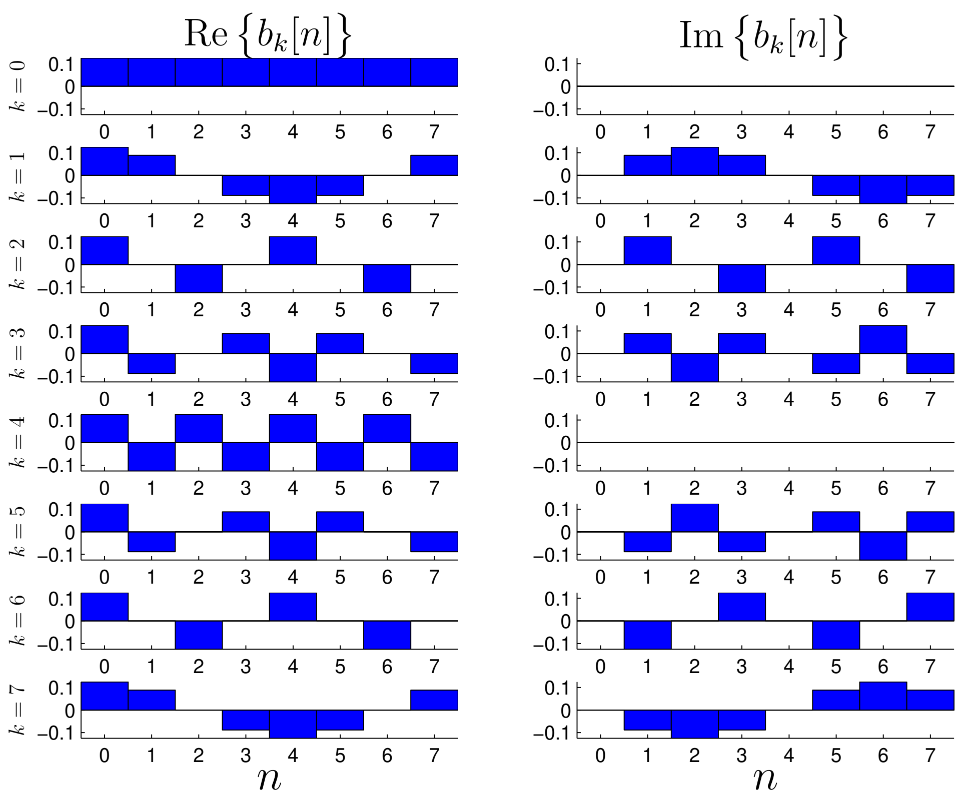
Recall from IDFT that $$x[n] = \sum_{k=0}^{N-1} X[k] \underbrace{\tfrac{1}{N} e^{j\frac{2\pi}{N}kn}}_{\triangleq\, b_k[n]}.$$

Thus, any $\{x[n]\}_{n=0}^{N-1}$ can be expressed as a weighted sum of N orthogonal *basis waveforms* $\{b_k[n]\}_{n=0}^{N-1}$.

$$\sum_{n=0}^{N-1} b_k^*[n] b_l[n] = \frac{1}{N} \cdot \frac{1}{N} \sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}(l-k)n} = \frac{1}{N}\delta[l-k]$$

These act as elementary "building blocks" for signals.

Example for $N = 8$:



Circular Convolution:

What results from multiplying two N-point DFTs?

$$\begin{aligned} X[k]W[k] &= \sum_{m=0}^{N-1} x[m] e^{-j\frac{2\pi}{N}mk} \sum_{l=0}^{N-1} w[l] e^{-j\frac{2\pi}{N}lk} \\ &= \sum_{m=0}^{N-1} x[m] \sum_{l=0}^{N-1} w[l] e^{-j\frac{2\pi}{N}(m+l)k}. \end{aligned}$$

Using the substitution

$$l = \langle n-m \rangle_N = n - m + qN \text{ for some } q \in \mathbb{Z}$$

we get

$$\begin{aligned} X[k]W[k] &= \sum_{m=0}^{N-1} x[m] \sum_{n=0}^{N-1} w[\langle n-m \rangle_N] \underbrace{e^{-j\frac{2\pi}{N}(m+n-m+qN)k}}_{e^{-j\frac{2\pi}{N}nk}} \\ &= \sum_{n=0}^{N-1} \left(\sum_{m=0}^{N-1} x[m] w[\langle n-m \rangle_N] \right) e^{-j\frac{2\pi}{N}nk} \\ &\overset{N\text{-DFT}}{\longleftrightarrow} \sum_{m=0}^{N-1} x[m] w[\langle n-m \rangle_N] \triangleq \{x[n]\} \, \text{Ⓝ} \, \{w[n]\} \end{aligned}$$

known as "N-circular convolution."

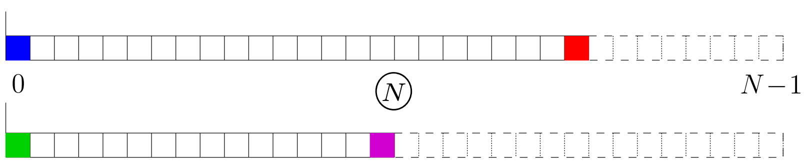
Circular Convolution (cont.):

Linear convolution: $$y[n] = \sum_{m=-\infty}^{\infty} x[m]w[n-m]:$$

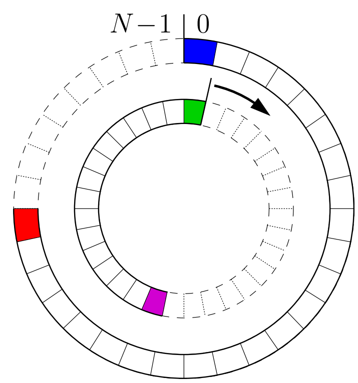
flip and slide (linearly). . .

$\boxed{N\text{-circular convolution:}}$ $$y[n] = \sum_{m=0}^{N-1} x[m]w[\langle n-m \rangle_N]:$$

flip and slide (cyclically). . .

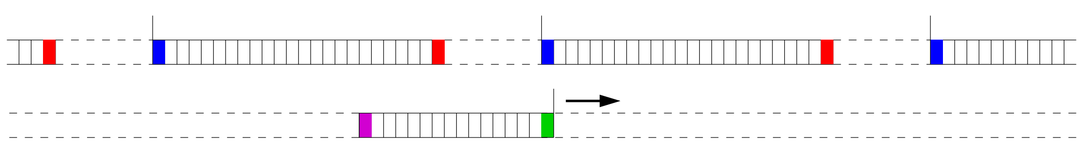


$=$



or cyclically extend, flip, and slide (linearly). . .

$=$



$$y[n] = \sum_{m=-\infty}^{\infty} x[\langle m \rangle_N]w[n-m]$$

(Assume finite-length $\{w[m]\}$ is zero-extended in last equation.)

Circular Convolution — Examples:

1. Consider $[1,2,3] \circledN [2,-1,1]$ for $N=3$:

1	2	3	1	2	3	1	2	3	out:
	1	-1	2						1
		1	-1	2					6
			1	-1	2				5
				1	-1	2			1
					1	-1	2		6
						1	-1	2	5
							$\ddots$		$\vdots$

2. Now consider $[1,2,3] \circledN [2,-1,1]$ for $N=5$,
   or, equivalently, $[1,2,3,0,0] \circledN [2,-1,1,0,0]$ for $N=5$:

1	2	3	0	0	1	2	3	0	0	out:
	0	0	1	-1	2					2
		0	0	1	-1	2				3
			0	0	1	-1	2			5
				0	0	1	-1	2		-1
					0	0	1	-1	2	3
						$\ddots$				$\vdots$

The DFT — important properties:

Here, $x[n] \overset{N\text{-DFT}}{\longleftrightarrow} X[k]$, $w[n] \overset{N\text{-DFT}}{\longleftrightarrow} W[k]$, and $n,k = 0 \ldots N-1$.

1. Linearity:

$$\alpha x[n] + \beta w[n] \;\overset{N\text{-DFT}}{\longleftrightarrow}\; \alpha X[k] + \beta W[k]$$

2. Conjugate symmetry:

$$x^*[n] = x[\langle -n\rangle_N] \;\Leftrightarrow\; X[k] \in \mathbb{R}$$

$$x[n] \in \mathbb{R} \;\Leftrightarrow\; X^*[k] = X[\langle -k\rangle_N]$$

3. Shift/modulation:

$$x[\langle n-d\rangle_N] \;\overset{N\text{-DFT}}{\longleftrightarrow}\; e^{-j\frac{2\pi}{N}dk} X[k],\; d \in \mathbb{Z}$$

$$e^{j\frac{2\pi}{N}dn} x[n] \;\overset{N\text{-DFT}}{\longleftrightarrow}\; X[\langle k-d\rangle_N],\; d \in \mathbb{Z}$$

4. Reversal:

$$x[\langle -n\rangle_N] \;\overset{N\text{-DFT}}{\longleftrightarrow}\; X[\langle -k\rangle_N]$$

5. Multiplication/convolution:

$$x[n]w[n] \;\overset{N\text{-DFT}}{\longleftrightarrow}\; \frac{1}{N}\sum_{l=0}^{N-1} X[l]W[\langle k-l\rangle_N]$$

$$\sum_{m=0}^{N-1} x[m]w[\langle n-m\rangle_N] \;\overset{N\text{-DFT}}{\longleftrightarrow}\; X[k]W[k]$$

6. Energy conservation ("Parseval's theorem"):

$$\sum_{n=0}^{N-1} |x[n]|^2 \;=\; \frac{1}{N}\sum_{k=0}^{N-1} |X[k]|^2$$

Working with modulo indices:

- Many DFT properties involve modulo-N indices!
- *Main idea:* $\boxed{\langle m\rangle_N = m + qN \text{ for integer } q}$ (in fact $q = -\lfloor \frac{m}{N} \rfloor$, but the exact value will not matter). Recall circular convolution example (p. 7). Here's another...
- <u>Proof of $x[\langle -n\rangle_N] \overset{N\text{-DFT}}{\longleftrightarrow} X[\langle -k\rangle_N]$:</u>

$$x[\langle -n\rangle_N] \overset{N\text{-DFT}}{\longleftrightarrow} \sum_{n=0}^{N-1} x[\underbrace{\langle -n\rangle_N}_{-n+qN \triangleq m}]e^{-j\frac{2\pi}{N}nk} \quad \text{for integer } q$$

$$= \sum_{m=0}^{N-1} x[m]e^{-j\frac{2\pi}{N}(qN-m)k} \quad \text{same sum!}$$

$$= \sum_{m=0}^{N-1} x[m]e^{-j\frac{2\pi}{N}m(-k)}$$

Can't write "$X[-k]$" because DFT is only defined for $k = 0 \ldots N-1$! Instead, $\langle -k\rangle_N = -k + rN$ for $r \in \mathbb{Z}$:

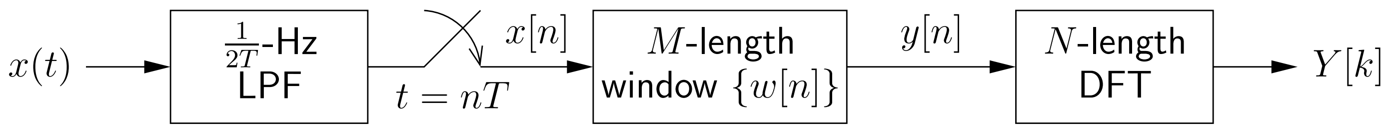
$$= \sum_{m=0}^{N-1} x[m]e^{-j\frac{2\pi}{N}m(\langle -k\rangle_N - rN)} \quad \text{for integer } r$$

$$= X[\langle -k\rangle_N]$$

- Example:

n :	-5	-4	-3	-2	-1	0	1	2	3	4	5
$x[n]$:						a	b	c	d	e	
$x[-n]$:		e	d	c	b	a					
$x[\langle -n\rangle_5]$:	a	e	d	c	b	a	e	d	c	b	a

Spectral Analysis:



Goal: Characterize the sinusoidal components in $x(t)$.

Method: Sample, window (i.e., $y[n] = w[n]x[n]$), and transform non-zero samples to the DFT domain.

Questions: How to choose M, $\{w[n]\}$, and N?
How to interpret DFT outputs $\{Y[k]\}$?

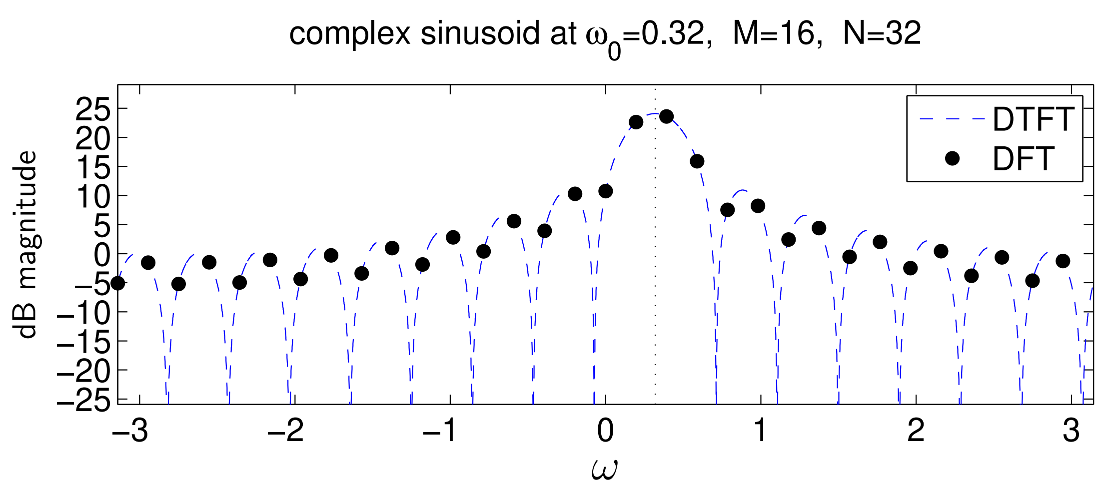
Recall key relationships:

$$X(e^{j\omega}) = \frac{1}{T} X_c(j\tfrac{\omega}{T}), \quad \omega \in [-\pi, \pi) \qquad \text{(no aliasing)}$$

$$Y(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\xi}) W(e^{j(\omega-\xi)}) d\xi \qquad (y[n] = w[n]x[n])$$

$$Y[k] = Y(e^{j\frac{2\pi}{N}k}) \qquad \text{(if } N \geq M\text{)}$$

Example:



Spectral Analysis (cont.):

Suppose our signal is a sum of L complex sinusoids:

$$x(t) = \sum_{l=1}^{L} a_l e^{j\Omega_l t} \quad \text{for } |\Omega_l| < \tfrac{\pi}{T}.$$

Then, using the relationships from the previous page,

$$\begin{aligned}
X_\mathsf{c}(j\Omega) &= 2\pi \sum_{l=1}^{L} a_l \delta(\Omega - \Omega_l) \\
X(e^{j\omega}) &= \frac{2\pi}{T} \sum_{l=1}^{L} a_l \delta(\tfrac{\omega}{T} - \Omega_l) \;\; \text{for } \omega \in [-\pi, \pi) \\
Y(e^{j\omega}) &= \frac{1}{T} \int_{-\pi}^{\pi} \sum_{l=1}^{L} a_l \delta(\tfrac{\xi}{T} - \Omega_l) W(e^{j(\omega-\xi)}) d\xi \\
&= \sum_{l=1}^{L} a_l \int_{-\pi/T}^{\pi/T} \delta(\theta - \Omega_l) W(e^{j(\omega-\theta T)}) d\theta \text{ via } \theta = \tfrac{\xi}{T} \\
&= \sum_{l=1}^{L} a_l W(e^{j(\omega-\Omega_l T)})
\end{aligned}$$

. . . a sum of scaled and shifted window-DTFTs! Finally,

$$Y[k] = \sum_{l=1}^{L} a_l W(e^{j(\frac{2\pi}{N}k - \Omega_l T)})$$

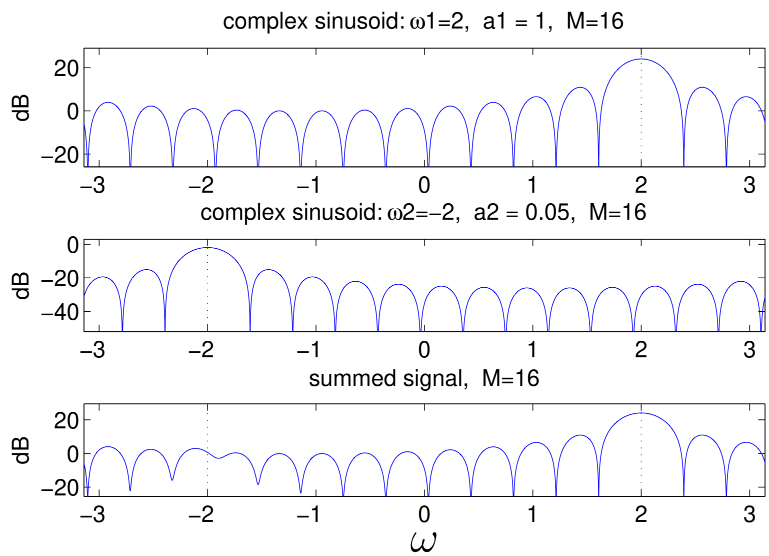
So we expect a DFT peak near $\boxed{k \approx \frac{\Omega_l T N}{2\pi}}$ for each $l = 1 \ldots L$.

Effects of finite data-length:

With $L > 1$ complex sinusoids, the DFT peaks may not be where we expect! There are two main reasons for this:

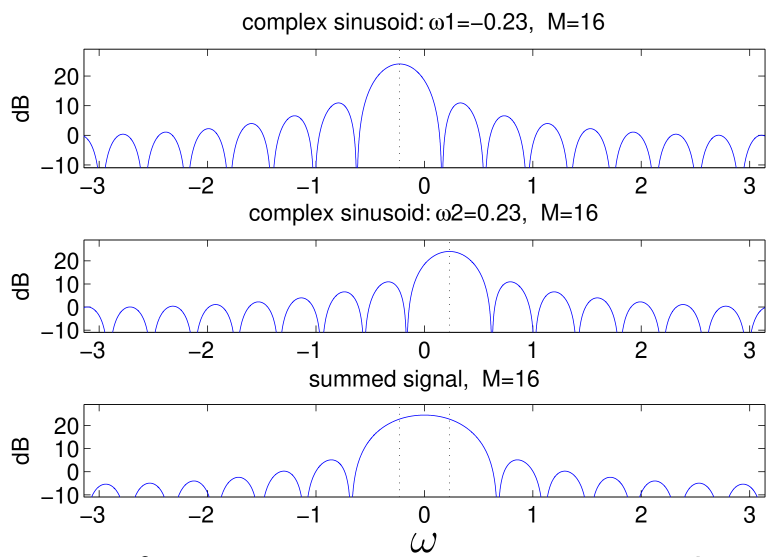
1. Leakage: The energy at $\omega_l = \Omega_l T$ gets spread over all $\omega \in [-\pi, \pi]$ through the window's sidelobes.

Weak sinusoids get buried!



2. Limited Resolution: The energy at $\omega_l = \Omega_l T$ gets spread broadly over the window's main-lobe.

Closely spaced sinusoids become indistinguishable!



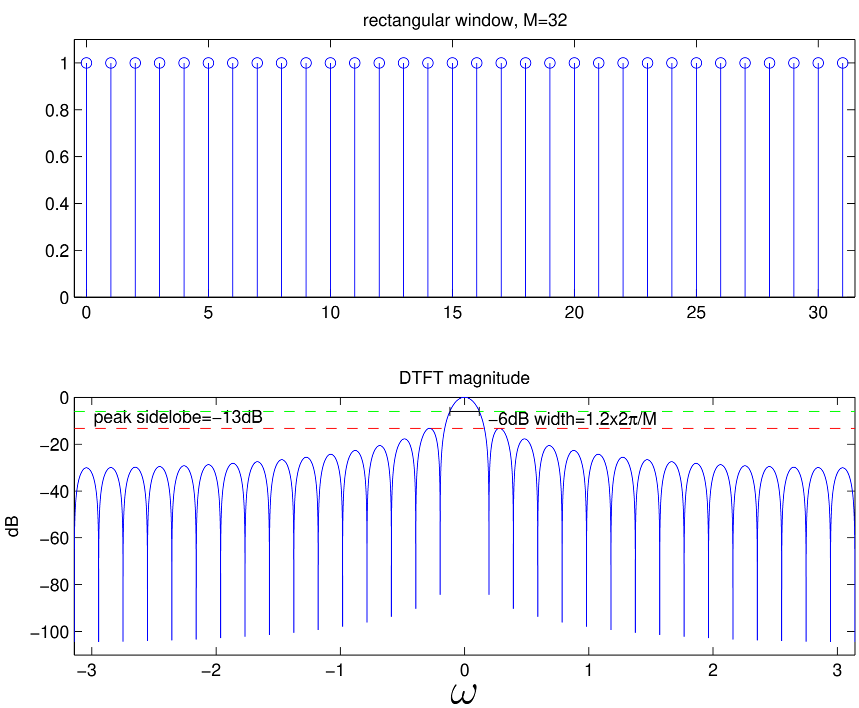
Can **trade** between leakage & resolution via choice of window.

Rectangular window:

Recall that, for an M-length rectangular window, we saw

$$W(e^{j\omega}) = \frac{\sin(\omega M/2)}{\sin(\omega/2)} e^{-j\omega(M-1)/2}.$$

So, in time- and frequency-domain, this window looks like:

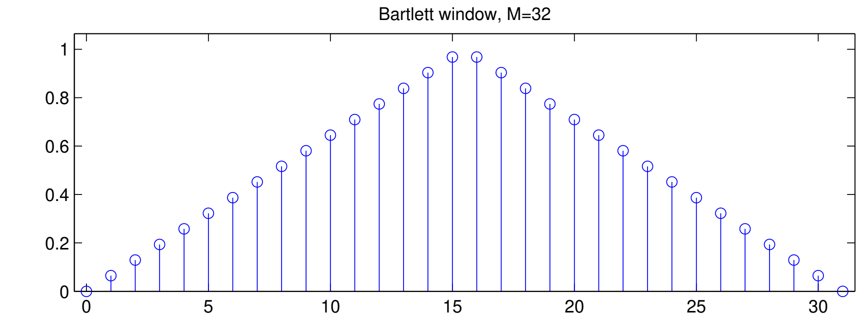


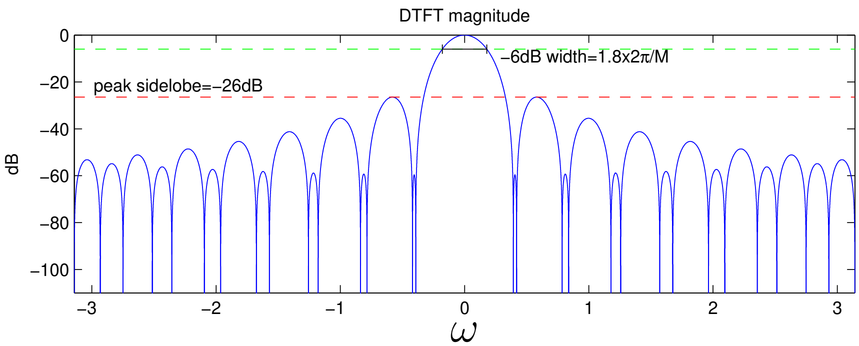
peak sidelobes @ -13dB

resolution limit $\approx 1.2 \times \frac{2\pi}{M}$

where "resolution limit" is "-6 dB bandwidth" and approximates the minimum spacing between recognizable sinusoids (of equal amplitude).

Bartlett window:

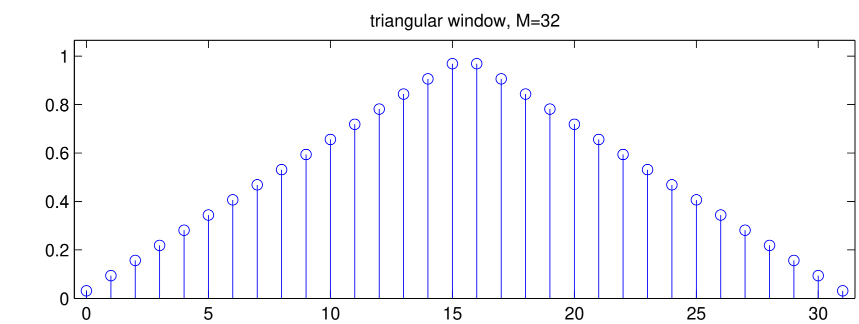




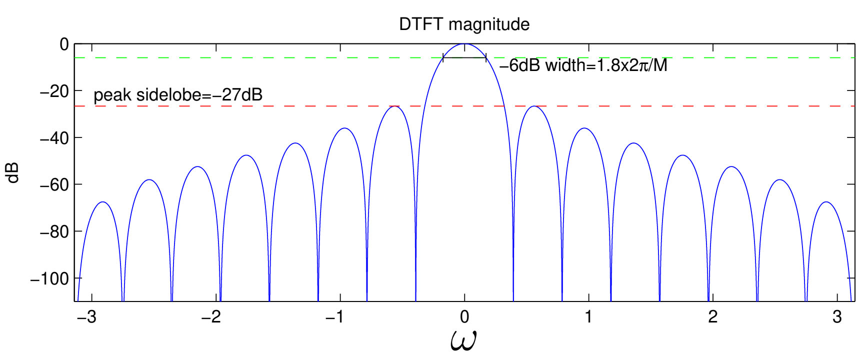
peak sidelobes @ -26dB

resolution limit $\approx 1.8 \times \frac{2\pi}{M}$

Triangular window:



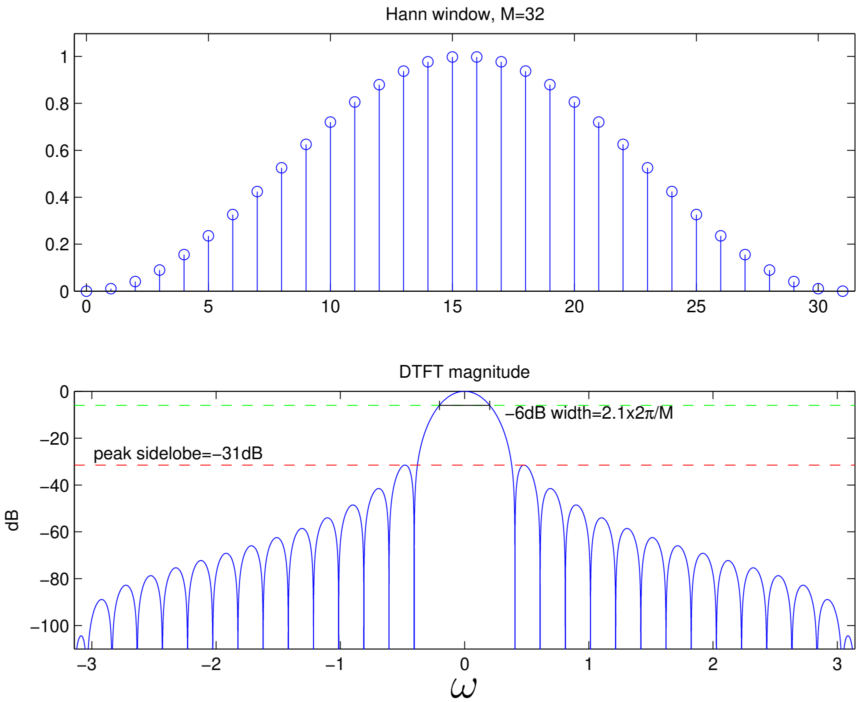
improved: edge points are not zero-valued!



peak sidelobes @ -27dB

resolution limit $\approx 1.8 \times \frac{2\pi}{M}$

Hann window:

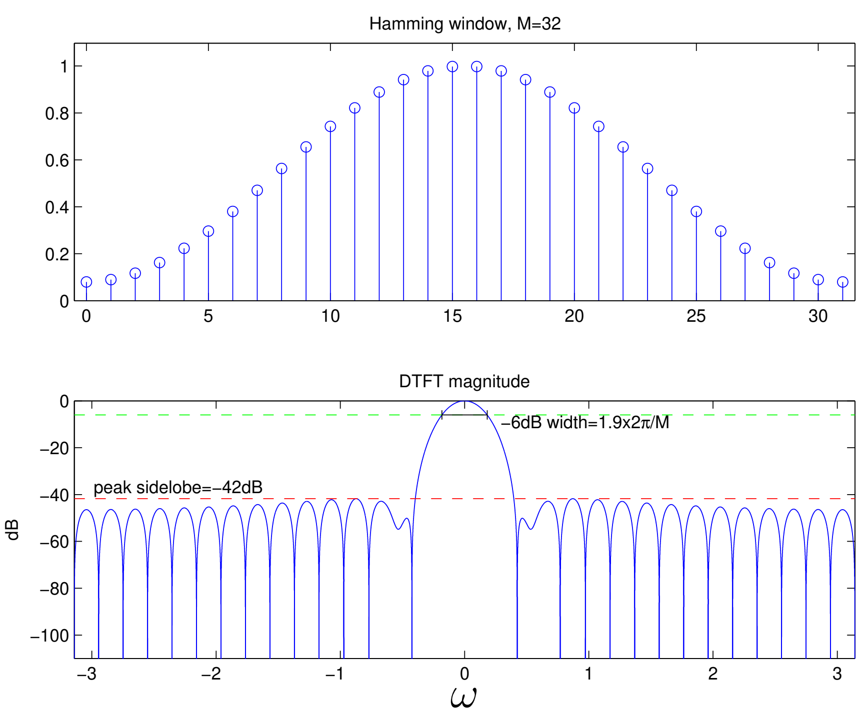


$w[n] = \frac{1}{2} - \frac{1}{2}\cos(\frac{2\pi}{M}n)$

peak sidelobes @ -31dB

resolution limit $\approx 2.1 \times \frac{2\pi}{M}$

Hamming window:

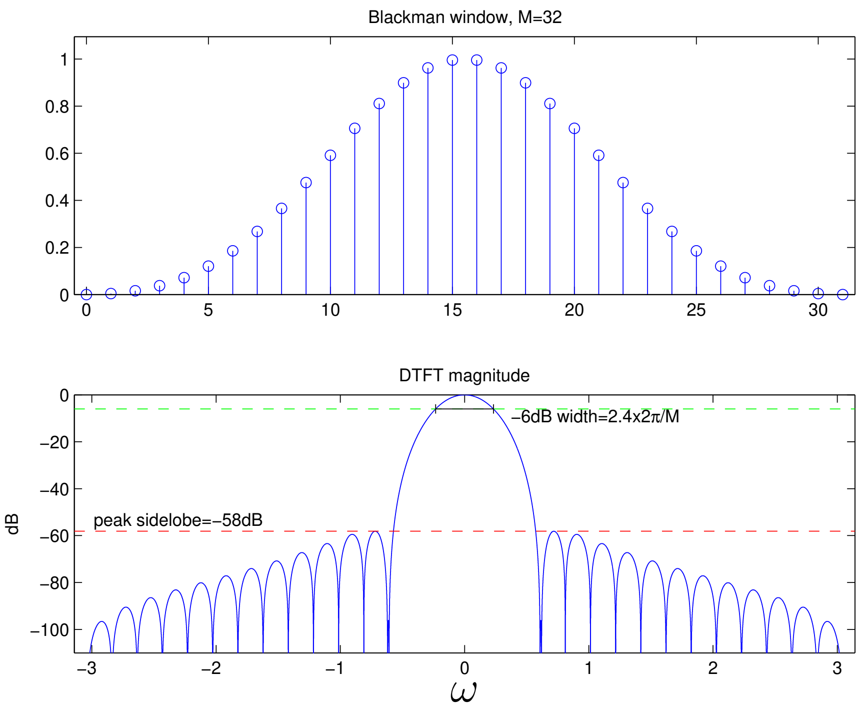


$w[n] = 0.538 - 0.462\cos(\frac{2\pi}{M}n)$

peak sidelobes @ -42dB

resolution limit $\approx 1.9 \times \frac{2\pi}{M}$

Blackman window:

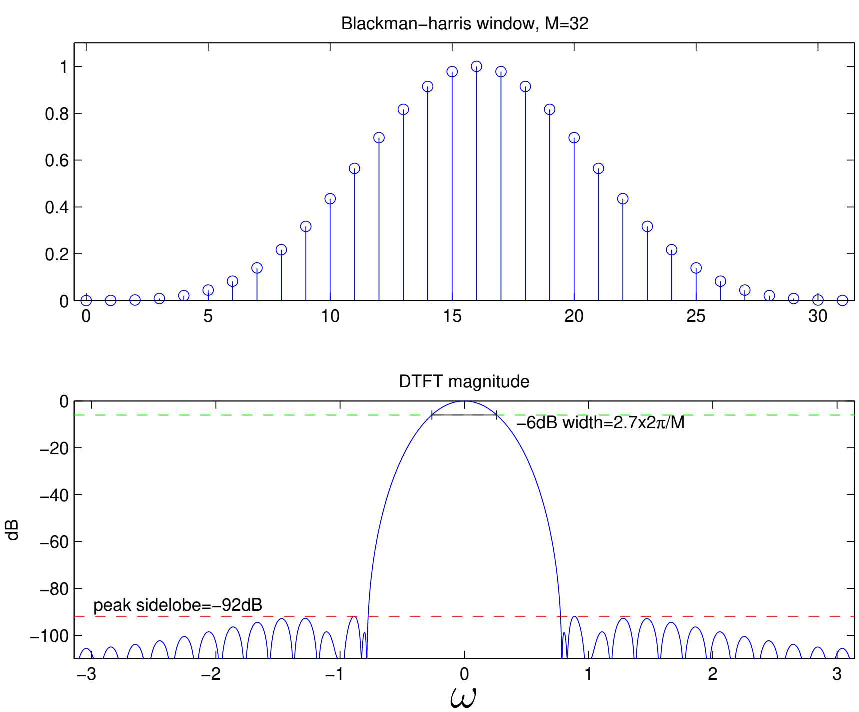


$w[n] = 0.42 - \frac{1}{2}\cos(\frac{2\pi}{M}n) + 0.08\cos(\frac{4\pi}{M}n)$

peak sidelobes @ -58dB

resolution limit $\approx 2.4 \times \frac{2\pi}{M}$

Blackman-harris window:



optimized 4-term

peak sidelobes @ -92dB

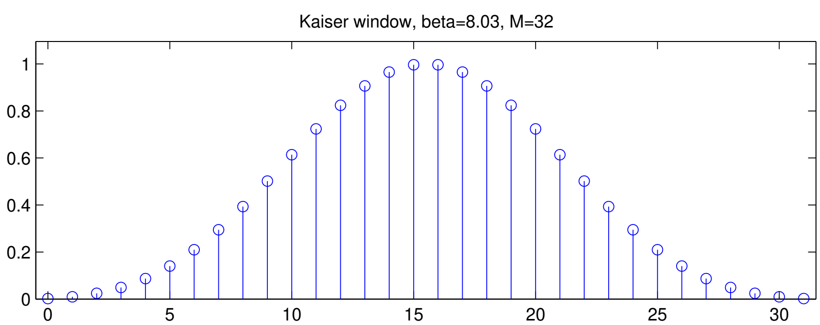
resolution limit $\approx 2.7 \times \frac{2\pi}{M}$

Kaiser window:

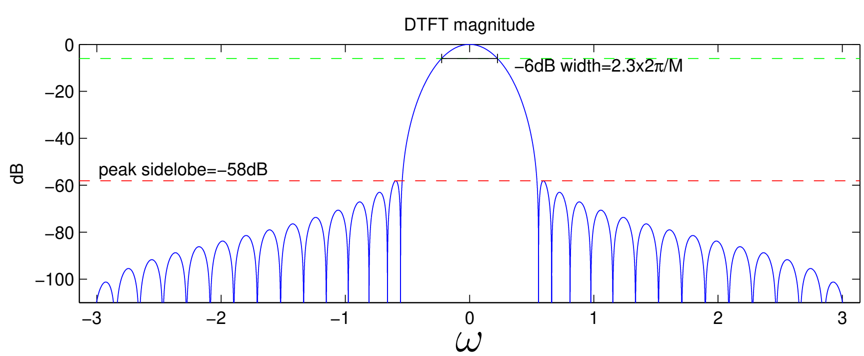
Has adjustable parameter β. For peak sidelobe level $-S_{\text{dB}}$,

$$\beta = \begin{cases} 0 & S_{\text{dB}} = 13.26 \\ 0.76609(S_{\text{dB}} - 13.26)^{0.4} + 0.09834(S_{\text{dB}} - 13.26) & 13.26 < S_{\text{dB}} \leq 60 \\ 0.12438(S_{\text{dB}} + 6.3) & 60 < S_{\text{dB}} \leq 120 \end{cases}$$

For example, $\beta = 8.03$ gives $S_{\text{dB}} = 59$ and, when $M = 32$,



based on modified Bessel function



peak sidelobes @ -58dB

resolution limit $\approx 2.3 \times \frac{2\pi}{M}$

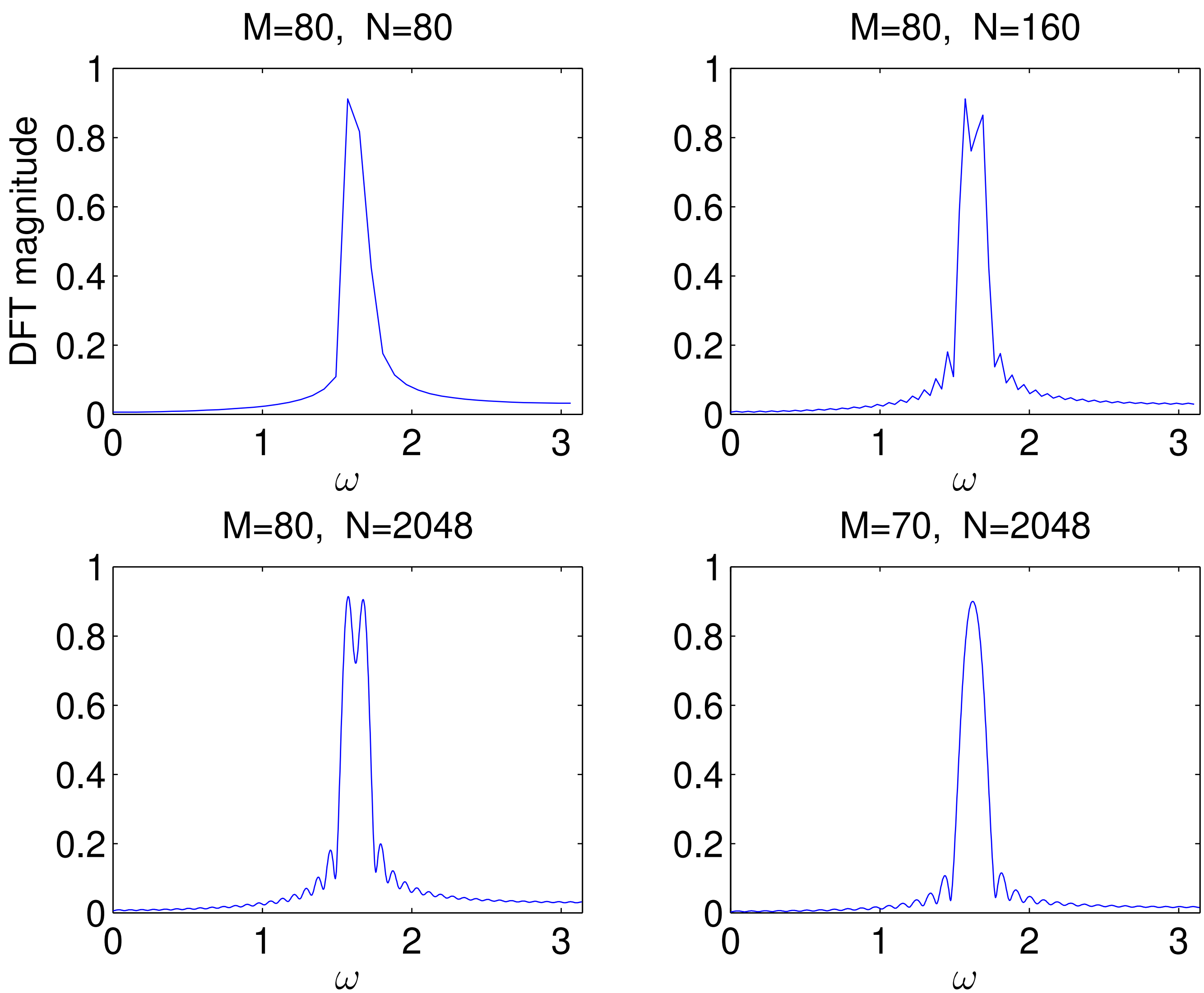
which is very similar (and slightly better) than Blackman.

The approximate tradeoff between peak sidelobe level $-S_{\text{dB}}$ and -6dB mainlobe width Δ_ω (in rad/sample) is given by

$$\Delta_\omega \;\approx\; 0.029(29 + S_{\text{dB}} - 0.003\, S_{\text{dB}}^2)\frac{2\pi}{M}.$$

Spectral analysis example:

- Real sines at 30 Hz & 32 Hz, $\frac{1}{T} = 120$ Hz. $(\Delta\omega = \frac{2\times 2\pi}{120})$
- Recall: M-length data, N-length DFT.
- Rectangular window: resolution limit $\frac{1.2\times 2\pi}{M}$ $(\Rightarrow M > 72)$.



- In general:

$$\boxed{\text{Need DFT length } N \geq 2M\text{, but prefer } N \gg 2M.}$$

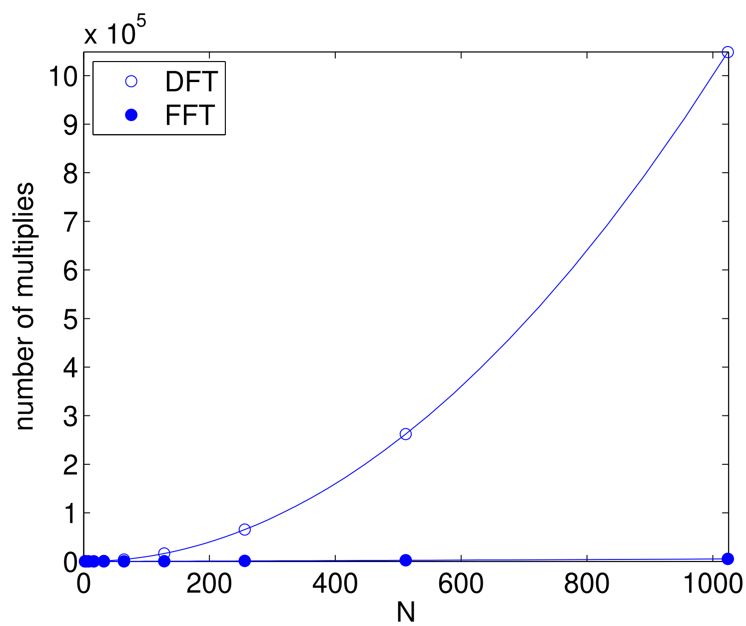
The fast Fourier transform (FFT):

- A family of algorithms to quickly compute the DFT.
- Recall that the N-DFT

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

  consumes N (complex-valued) multiplications per output, and thus N^2 multiplications total.

- We will describe the so-called "radix-2 decimate-in-time" FFT, which works when N is power-of-two, and consumes approximately $\frac{N}{2}\log_2 N$ multiplications.



For large N, the FFT is much cheaper than the brute-force DFT!

The radix-2 FFT:

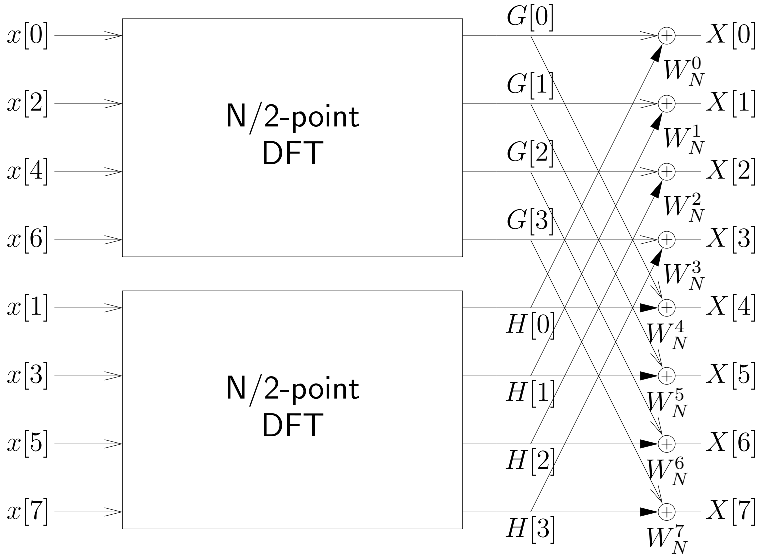
Notice that

$$\begin{aligned} X[k] &= \sum_{n=0}^{N-1} x[n] W_N^{kn} \qquad \text{for} \;\; W_N^q \triangleq e^{-j\frac{2\pi}{N}q} \\ &= \sum_{n \text{ even}} x[n] W_N^{kn} + \sum_{n \text{ odd}} x[n] W_N^{kn} \\ &= \sum_{m=0}^{N/2-1} x[2m] W_N^{k2m} + \sum_{m=0}^{N/2-1} x[2m+1] W_N^{k(2m+1)} \end{aligned}$$

Because $W_N^{2q} = W_{N/2}^q$ and $W_N^{q+p} = W_N^q W_N^p$, we can write

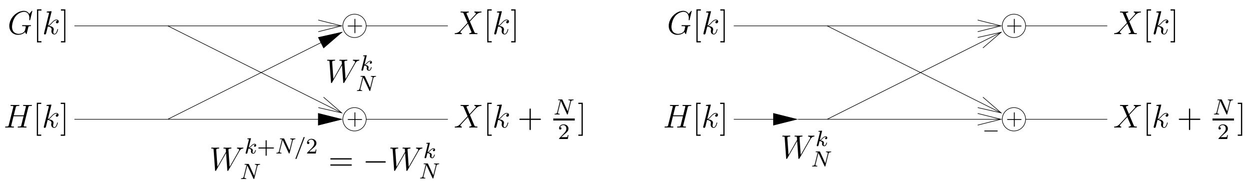
$$X[k] = \underbrace{\sum_{m=0}^{N/2-1} x[2m] W_{N/2}^{km}}_{\triangleq\, G[k]} + W_N^k \underbrace{\sum_{m=0}^{N/2-1} x[2m+1] W_{N/2}^{km}}_{\triangleq\, H[k]}$$

Note: $G[k]$ is an $\frac{N}{2}$-DFT of the even samples, and $H[k]$ is an $\frac{N}{2}$-DFT of the odd samples. Both are $\frac{N}{2}$-periodic in k.



The radix-2 FFT (cont.):

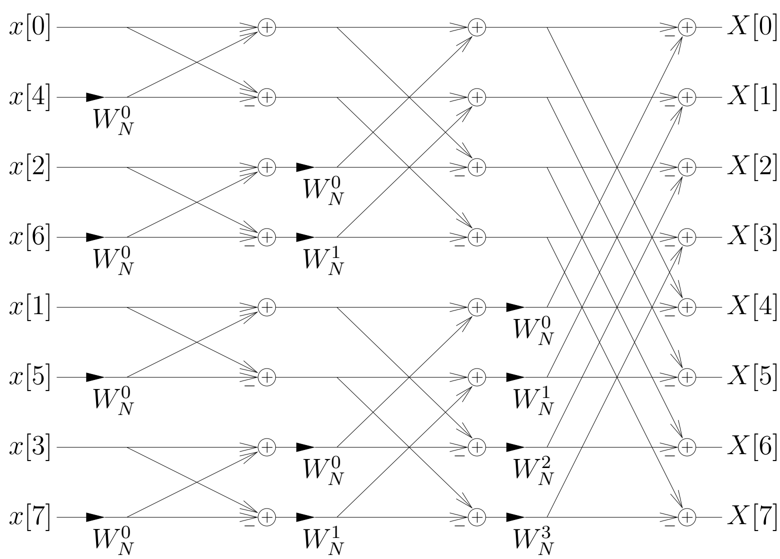
- The final processing stage can be simplified from N mults to $\frac{N}{2}$ mults because the following are equivalent:



- Let's examine the complexity of this split-in-half method:
  - direct N-DFT: N^2 mults,
  - split-in-half: $(\frac{N}{2})^2 + (\frac{N}{2})^2 + \frac{N}{2} = \frac{1}{2}(N^2 + N)$ mults,

  Since $\frac{1}{2}(N^2 + N) < N^2$ for any $N > 1$, *it pays to split!*

- Thus, we split each $\frac{N}{2}$-DFT into two $\frac{N}{4}$-DFTs (combining their outputs using twiddle factors), repeat with each $\frac{N}{4}$-DFT, and so on, finally obtaining

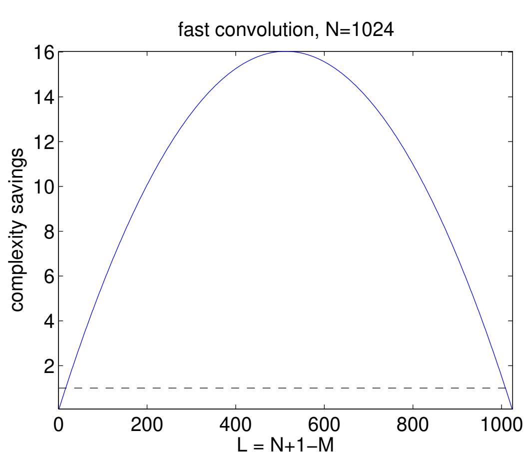


In total, $(\log_2 N)$ stages $\times$ $\frac{N}{2}\frac{\text{mults}}{\text{stage}} = \frac{N}{2}\log_2 N$ mults.

Fast convolution:

The FFT gives us a very fast way to do linear convolution.

- Say that we want to *linearly* convolve an M-length signal $x[n]$ with an L-length signal $h[n]$, i.e.,
$$y[n] = \sum_{l=0}^{L-1} h[l]x[n-l],$$
requiring $\approx L$ mults per output, or ML mults in total.
- If we zero-pad both sequences to length $N \geq M + L - 1$, then N-*circular* convolution yields the same outputs as linear convolution. (Recall page 11.)
- Furthermore, N-*circular* convolution can be implemented as multiplication in the N-DFT domain:
$$x[n] \circledN h[n] = \mathcal{F}^{-1}_{N\text{-DFT}}\Big\{\mathcal{F}_{N\text{-DFT}}\{x[n]\} \cdot \mathcal{F}_{N\text{-DFT}}\{h[n]\}\Big\}$$
and, if we use FFT/IFFT to implement the DFT/IDFT, this requires only $3 \times \frac{N}{2}\log_2 N + N$ total mults.



Much cheaper unless M or L very small!

Overlap-Save:

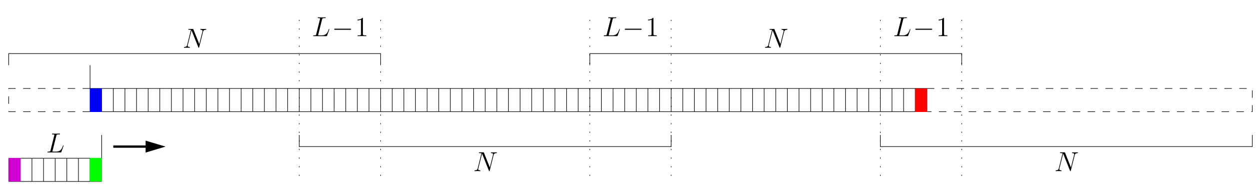
- Say that we want to filter M-length signal $x[n]$ with L-length filter impulse response $h[n]$.
- Say that we have access to an N-length FFT algorithm, where $L < N < M + L - 1$.

  Rather than zero-padding $x[n]$, we break it up into overlapping segments of length N, and perform fast convolution on each one.

- Key quantities:

  $M+L-1 =$ # linear-convolution outputs to compute
  $N-L+1 =$ # linear-conv outputs per N-circular conv
  $\lceil \frac{M+L-1}{N-L+1} \rceil =$ # N-circular convolutions needed in total

  Example with $L = 8$, $N = 32$, $M = 72$:



- Total multiplies required for overlap-save:

$$\underbrace{\tfrac{N}{2}\log_2 N}_{\texttt{H=fft(h)}} + \lceil \tfrac{M+L-1}{N-L+1} \rceil \Big(\underbrace{\tfrac{N}{2}\log_2 N}_{\texttt{X=fft(x)}} + \underbrace{N}_{\texttt{Y=X.*H}} + \underbrace{\tfrac{N}{2}\log_2 N}_{\texttt{y=ifft(Y)}} \Big)$$

  Compare to ML multiplies for standard linear conv.

Overlap-Save Example:

- Say we want to compute the linear convolution $[a, b, c, d] * [1, 1]$ using several 3-circular convolutions.
- We know that the desired output is

$$[a,\ a{+}b,\ b{+}c,\ c{+}d,\ d]$$

- In this example, $M{=}4$, $L{=}2$, and $N{=}3$, which means:

  $M{+}L{-}1 = 5$ linear convolution outputs

  $N{-}L{+}1 = 2$ good outputs per N-circular convolution

  $\lceil \frac{M+L-1}{N-L+1} \rceil = 3$ N-circular convolutions

- We prepare the signal by inserting $L{-}1 = 1$ zeros in front and as many as needed in back.

$$0,\ a,\ b,\ c,\ d,\ 0,\ 0$$

  The N-blocks overlap by $L{-}1 = 1$ samples, as shown.

- The N-circular convolutions manifest as:

  $[0, a, b] \textcircled{3} [1, 1] = [\cancel{b},\ a,\ a{+}b]$

  $[b, c, d] \textcircled{3} [1, 1] = [\cancel{d{+}b},\ b{+}c,\ c{+}d]$

  $[d, 0, 0] \textcircled{3} [1, 1] = [\cancel{d},\ d,\ 0]$

  and we discard the first $L{-}1 = 1$ samples of each.

- Concatenating the good circular-convolution outputs gives

$$[a,\ a{+}b,\ b{+}c,\ c{+}d,\ d,\ 0]$$

  which is the desired result (plus some zero padding).

Matrix/vector formulations:

Many of the properties we've seen so far can be efficiently described using matrices and vectors.

- The N-DFT $X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$ can be written in matrix/vector form as

$$\underbrace{\begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[N-1] \end{bmatrix}}_{\boldsymbol{X}} = \underbrace{\begin{bmatrix} e^{-j\frac{2\pi}{N}\cdot 0\cdot 0} & e^{-j\frac{2\pi}{N}\cdot 0\cdot 1} & \cdots & e^{-j\frac{2\pi}{N}\cdot 0\cdot (N-1)} \\ e^{-j\frac{2\pi}{N}\cdot 1\cdot 0} & e^{-j\frac{2\pi}{N}\cdot 1\cdot 1} & \cdots & e^{-j\frac{2\pi}{N}\cdot 1\cdot (N-1)} \\ \vdots & \vdots & \ddots & \vdots \\ e^{-j\frac{2\pi}{N}\cdot (N-1)\cdot 0} & e^{-j\frac{2\pi}{N}\cdot (N-1)\cdot 1} & \cdots & e^{-j\frac{2\pi}{N}\cdot (N-1)\cdot (N-1)} \end{bmatrix}}_{\boldsymbol{W}} \underbrace{\begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix}}_{\boldsymbol{x}}.$$

  The k^{th} row & n^{th} column of DFT matrix $\boldsymbol{W}$ is $e^{-j\frac{2\pi}{N}kn}$.

- The IDFT matrix is $\boldsymbol{W}^{-1} = \frac{1}{N}\boldsymbol{W}^{\mathsf{H}}$, where $(\cdot)^{\mathsf{H}}$ denotes the "Hermitian," i.e., conjugate transpose. Notice

$$\tfrac{1}{N}\boldsymbol{W}^{\mathsf{H}}\boldsymbol{W} = \boldsymbol{I} = \tfrac{1}{N}\boldsymbol{W}\boldsymbol{W}^{\mathsf{H}},$$

  since

$$\begin{aligned} \left[\tfrac{1}{N}\boldsymbol{W}\boldsymbol{W}^{\mathsf{H}}\right]_{k,l} &= \frac{1}{N}\sum_{n=0}^{N-1} e^{-j\frac{2\pi}{N}kn} e^{j\frac{2\pi}{N}ln} \\ &= \frac{1}{N}\sum_{n=0}^{N-1} e^{-j\frac{2\pi}{N}n(k-l)} \\ &= \begin{cases} 1 & k=l \\ 0 & k\neq l \end{cases} = \left[\boldsymbol{I}\right]_{k,l}. \end{aligned}$$

Matrix/vector formulations (cont.):

- An important class of matrices are "unitary" matrices, i.e., matrices $\boldsymbol{U}$ such that

$$\boldsymbol{U}^{\mathsf{H}}\boldsymbol{U} = \boldsymbol{I} = \boldsymbol{U}\boldsymbol{U}^{\mathsf{H}}.$$

  Note that the normalized DFT matrix $\frac{1}{\sqrt{N}}\boldsymbol{W}$ is unitary.

- Unitary matrices implement *rotations*: $\boldsymbol{U}\boldsymbol{x}$ is a rotation of $\boldsymbol{x} \in \mathbb{C}^N$ in the space $\mathbb{C}^N$. Essentially, the (scaled) DFT $\frac{1}{\sqrt{N}}\boldsymbol{W}$ rotates the time-domain signal vector $\boldsymbol{x}$ so that its Fourier components (i.e., complex exponentials) are visible. Sometimes, this is a better "view" of the signal.

- Notice that rotating a vector does not change its length, as measured by the standard Euclidean norm

$$\|\boldsymbol{x}\|_2 \triangleq \sqrt{\sum_{n=0}^{N-1} |x[n]|^2}.$$

  In other words, $\|\boldsymbol{U}\boldsymbol{x}\|_2 = \|\boldsymbol{x}\|_2$ for any unitary $\boldsymbol{U}$. This explains Parseval's theorem for the DFT:

$$\sum_{n=0}^{N-1} |x[n]|^2 = \|\boldsymbol{x}\|_2^2 = \|\underbrace{\tfrac{1}{\sqrt{N}}\boldsymbol{W}}_{\boldsymbol{U}}\boldsymbol{x}\|_2^2 = \tfrac{1}{N}\|\boldsymbol{W}\boldsymbol{x}\|_2^2 = \frac{1}{N}\sum_{k=0}^{N-1} |X[k]|^2.$$

Matrix/vector formulations (cont.):

- Linear convolution $y[n] = \sum_{l=-\infty}^{\infty} x[l]h[n-l]$ can be written as, for example,

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \\ y[4] \\ y[5] \\ y[6] \end{bmatrix} = \begin{bmatrix} h[0] & 0 & 0 & 0 \\ h[1] & h[0] & 0 & 0 \\ h[2] & h[1] & h[0] & 0 \\ h[3] & h[2] & h[1] & h[0] \\ 0 & h[3] & h[2] & h[1] \\ 0 & 0 & h[3] & h[2] \\ 0 & 0 & 0 & h[3] \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix},$$

  while N-circular convolution $y[n] = \sum_{l=0}^{N-1} x[l]h[\langle n-l \rangle_N]$ can be written as

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \end{bmatrix} = \begin{bmatrix} h[0] & h[3] & h[2] & h[1] \\ h[1] & h[0] & h[3] & h[2] \\ h[2] & h[1] & h[0] & h[3] \\ h[3] & h[2] & h[1] & h[0] \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix},$$

  where the matrix has a "circulant" structure.

- Circulant $\boldsymbol{H}$ with first column $\boldsymbol{h}$ has eigendecomposition

$$\boldsymbol{H} = \tfrac{1}{N}\boldsymbol{W}^{\mathsf{H}}\boldsymbol{D}\,\boldsymbol{W} \quad \text{where} \quad \boldsymbol{D} = \operatorname{diag}(\boldsymbol{W}\boldsymbol{h}).$$

  Thus, circular convolution $\boldsymbol{y} = \boldsymbol{H}\boldsymbol{x}$ can be written as

$$\boldsymbol{y} = \tfrac{1}{N}\boldsymbol{W}^{\mathsf{H}}\boldsymbol{D}\,\boldsymbol{W}\boldsymbol{x}$$

  which can be recognized as DFT-domain multiplication!